CSC 2510 002 Final Project Phase 1

RedDragon

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Agenda:

- Scope and Definition
- Collaboration and tools
- Roles
- Delivery Timeline
- Development Approaches
- CI/CD Approach

Scope and Definition

We aim to create a bash script that performs the following tasks:

1. Read in data from a provided website
2. Create users on the Ubuntu target system
3. Provide a manual add user function
4. Implement a licensing feature and provide a simple menu

Team Member Roles

Developer and Lead Auditor

Mallory Flesher will function as a Developer and **Lead Auditor**.

- Inspect documentation
- Assist in correcting / clarifying documentation and processes
- Report on the quality of processes and documentation
- Responsible for *Add User Accounts*

Development Approaches

Developer and Test Director

Zach Seitner will function as a Developer and **Test Director**.

- Develop comprehensive tests
- Communicate with teammates for desired outcomes
- Schedule and report on tests
- Responsible for *Read in Data from the provided website*

Team Member Roles

Scrum Master

Rachel Robertson will function as the **Scrum Master**.

- Project deadlines
- Speak on behalf of the team
- Assist in resolving issues
- Responsible for *Add User*, *Licensing*, *main function*

Development Approaches

Add User Function

Add Users Manually

- Option on the primary script alongside the automation option.
- Adds **inputs** from the user to a **text file**.
- Leverages the **add user accounts script** to complete account creation

Scrubbing

Inputs will be validated using **regex** and will be tested for prompt injection

Development Approaches

Licensing and Main Script

Licensing

- AI will be used to generate an algorithm for encrypting and producing a license
- The license will be checked for before the script is permitted to run

Main Script

- Menu
- Directs the primary flow of the application

Development Approaches

Read in Data from the provided website

Read Data

- Use **Curl** to pull the data from the website
- Parse the data using **grep** and **regex**
- Output to a **text file**

Data Formatting

The txt file will be **organized into blocks** of user information and the rest will be discarded.

Development Approaches

Add User Accounts

Add user accounts

- **Read the data** from the text file in user blocks
- Use built in bash commands like **"useradd"** to create accounts
- Add checks to ensure that **duplicate users** are not added

Generating Password

Will develop an algorithm to create **unique passwords** for each account.

Collaboration and tools

Collaboration:

- **Weekly Meetings** to provide regular updates on progress and steer development
- **Github** for code and file sharing
- **Discord** for quick communication

Tools

- Ubuntu 24.04
- Virtual box 7.1.12
- BASH v5.2.21
- Git
- Brave AI Leo

Delivery Timeline

General Overview

Ideal Completion Dates:

- **April 12th**, fully operational
- **April 14th**, Additional auditing and documentation clean up
- **April 28th** Program is entirely polished, all unforeseen errors have been corrected

Potential Setbacks:

- Integration errors
- Bugs
- Event on April 25th may stall CI /CD for 25th-28th

Delivery Timeline

In-Depth pt.1

April 6th – April 10th

- Creation of Scripts
- Script Testing

April 11th

- Integration day
- **Submission of code**

April 13th,14th

- Auditing / Documentation clean up
- Additional testing

April 15th – April 28th

- CI / CD

CI/CD Approach

Continuous Integration and Development

- Gather **feedback** from client
- **Document** requests
- **Correct** errors
- **Add** desired features
- Test and report
- Repeat

Thank You